Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/28/2015
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	In vivo induction of chromosome instability and aberrant patterns of DNA methylation in hematopoietic stem/progenitor cells by heavy ions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/01/2011	**End Date:**	06/30/2015
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2015, per NSSC information and PI (Ed., 4/22/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Whorton, Elbert Ph.D. (Galveston National Laboratory)		
Grant/Contract No.:	NNX11AK91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Risk of the development of fatal cancer has been considered the dominant risk from exposure to heavy ions found in the space environment. Cancer of blood cells, in particular myeloid leukemia (ML), is one of the major health concerns from exposure to space radiation. However, the risks for developing cancer of blood cells after exposure to space radiation remain uncertain. To increase the reliability of predicting risk for developing cancer of blood cells, a much improved understanding of space radiation-induced changes in the target cells (stem cells and their progenitors) in living organisms for the induction of this particular type of cancer is critically important. To reduce uncertainties in assessing risks for developing cancer of blood cells from exposure to heavy ions found in space, a meaningful combination of biological endpoints highly relevant to the induction of cancer of blood cells must be studied at the dose range that is similar to what astronauts encounter in space. In vivo studies of humans are not possible; thus animal studies are critically important. In this project, we used the CBA/CaJ mouse as an experimental model to investigate the biological effects of heavy ions. To do this, we exposed mice to two types of heavy ions found in space and relevant to NASA missions, i.e., silicon and titanium. We determined the effects of these ions on the induction of damage at the genetic level by measuring the frequency and type of initial- and late-occurring chromosome instability (normally known as genomic instability). Such analyses were done in hematopoietic stem/progenitor cells (HSPCs) residing in the bone marrow (BM) of exposed mice collected at different times post-irradiation using a novel genome-wide multicolor fluorescence in situ hybridization (whole genome mFISH). The measurement of chromosome instability in HSPCs was coupled with the characterization of damage at the epigenetic level (i.e., the damage affecting gene function without a change in the sequence of DNA molecules) by determining the levels of global of DNA-methylation, i.e., 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC). Both chromosome instability and epigenetic events are highly associated with cancer development and relevant to risk assessment, but they have not been applied with in vivo exposure to space radiation with silicon or titanium ions. Our research is the first to use two in vivo endpoints relevant to cancer induction in the HSPCs that are the target of radiation-induced ML to evaluate the effects of space-radiation. Importantly, since there is increasing evidence of a link between inflammation and cancer, we investigated the potential induction of acute and chronic inflammation in HSPCs, the total population of bone marrow (BM) cells, and liver (one of the critical targets for radiation-induced cancer) collected from these exposed mice. It is clear that the resulting data are critically important for NASA in predicting health risks from exposure to space radiation during space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve our understanding of the molecular basis for the influence of space radiation on genomic instability and aberrant DNA-methylation profiles. Hence, the obtained data are critically important for understanding risks associated with space travel. Further, knowing the pathways associated with response to radiation exposure could be useful for planning countermeasure strategies to protect astronauts or space travelers in the space environment. Likewise, such knowledge can also have Earth benefits because a large number of people on Earth are exposed to radiation at varying doses and in many different ways. These include: individuals exposed to low levels of radiation in daily life (e.g., medical diagnosis or airport safety) or an unavoidable exposure to low-level background radiation; patients who receive high doses of radiation from radiation therapy; victims of nuclear terrorism; members of the armed forces; workers in the nuclear power industry; and people living in homes surrounding the power plants.
Task Progress:	We have completed the project. We exposed mice to 0, 0.1, 0.25, or 0.5 Gy of heavy ions found in the space environment (i.e., 300 MeV/n 28Si ions or 1 GeV/n 48Ti ions). Bone marrow cells were collected from groups of mice (5 mice in each treatment group) at various times, up to 6 months, post-irradiation for obtaining HSCP-derived clones by means of an in vitro colony forming unit assay. The highlights of our research are: • Heavy ions induced a persistent loss of 5hmC, chronic inflammation, and genomic instability (determined by the late-occurring chromosome aberrations in HSPCs). • A loss of global 5hmC is closely linked to genomic instability and chronic inflammation in HSPCs, total population of bone marrow cells, and liver tissues. • In contrast to global 5mC, a loss of global 5hmC holds promise as a putative molecular biomarker of radiation exposure. We provide for the first time the link among in vivo induction of chronic inflammation, aberrant pattern of DNA methylation (in particular a loss of 5hmC), and genomic instability in cells that are the targets of cancer induction after exposure to heavy ions found in the space environment. These biological endpoints are among the hallmarks of carcinogenesis. Hence, exposure to heavy ions in space may pose a risk of developing cancer, i.e., ML or liver cancer. The results also offer opportunities for developing new biological countermeasures to mitigate radiation-induced deleterious effects. These include anti-inflammatory agents or agents that can activate the activity of enzymes involved in an oxidation of 5mC to 5hmC.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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